

Quantifying Shadow Deviations of Bardeen Regular Black Holes: A Reproducible Computational Assessment for Event Horizon Telescope Observations

Aswin Karkadakattil

Independent Researcher in Advanced Manufacturing and Aerospace Engineering, with Applications in Artificial Intelligence, Kerala, India

Correspondence: ashwinharik20000@gmail.com

Abstract

The Kerr metric provides the standard description of astrophysical black holes, while singularity-free regular black-hole models offer alternative spacetime geometries that can be tested through horizon-scale observations. Here, we quantify the shadow deviations produced by the non-rotating Bardeen regular black-hole metric and assess their observational scale for Event Horizon Telescope (EHT) targets. A lightweight and fully reproducible computational framework is implemented in a single Google Colab notebook, using the photon-sphere condition and critical impact parameter to determine the Bardeen shadow boundary and its dependence on the regularisation parameter g . The calculation recovers the Schwarzschild shadow diameter of $10.3923M$ at $g/M=0$ and shows a smooth reduction to $10.2319M$, $9.6795M$, and $8.8817M$ at $g/M=0.3$, 0.6 , and 0.8 , corresponding to fractional deviations of approximately 1.54% , 6.86% , and 14.53% , respectively. For Sagittarius A*, the corresponding angular diameter decreases from $53.94\mu\text{as}$ at $g/M=0$ to $53.11\mu\text{as}$ at $g/M=0.3$ and $50.24\mu\text{as}$ at $g/M=0.6$; for M87*, it decreases from $39.69\mu\text{as}$ to $39.08\mu\text{as}$ and $36.97\mu\text{as}$, respectively. Across the tested range, both targets therefore remain above the adopted EHT angular-resolution benchmark, indicating that geometric resolvability is not equivalent to parameter distinguishability. The results provide a quantitative mapping between the Bardeen regularisation parameter and observable shadow-size deviations and establish a reproducible computational baseline for future comparisons incorporating black-hole spin, plasma effects, scattering, and full EHT image modelling. The complete workflow is designed for open reproduction and extension using standard scientific Python tools in a browser-based environment.

Keywords: black hole shadow; regular black holes; Bardeen metric; Event Horizon Telescope; null geodesics; ray tracing; Sagittarius A*; M87*; reproducible computational astrophysics

1. Introduction

General relativity describes astrophysical black holes through the Kerr metric, but regular black-hole models provide an alternative in which the central singularity is replaced by a nonsingular geometry. The Bardeen solution, originally proposed by Bardeen (1968), is one of the earliest and most widely studied examples. Ayón-Beato and García (1998, 1999, 2000) later demonstrated that the Bardeen geometry can arise as an exact solution of Einstein gravity coupled to nonlinear electrodynamics. Subsequent studies extended regular black holes to rotating configurations and investigated their photon trajectories and shadow properties (Bambi and Modesto, 2013; Li and Bambi, 2014; Abdujabbarov et al., 2016; Stuchlík and Schee, 2019; He et al., 2022). The development of horizon-scale very-long-baseline interferometry has made these theoretical differences increasingly relevant to observations. The Event Horizon Telescope (EHT) produced the first resolved image of M87* (Event Horizon Telescope Collaboration, 2019) and subsequently imaged Sagittarius A* (Sgr A*) at the centre of the Milky Way (Event Horizon Telescope Collaboration, 2022a). EHT observations have also been used to test the consistency of the observed image structure with the Kerr metric (Event Horizon Telescope Collaboration, 2022b). These results provide an observational basis for examining deviations from standard black-hole geometries through their shadow properties. Previous studies have established that regularisation can modify the photon sphere and shadow and that these effects can, in principle, be distinguished from Kerr spin (Li and Bambi, 2014; Abdujabbarov et al., 2016; Stuchlík and Schee, 2019). The influence of accretion and surrounding matter on Bardeen shadow appearance has also been investigated (He et al., 2022). However, an important practical question remains: **how does the Bardeen regularisation parameter translate into a measurable shadow-size deviation, and what angular scale does this deviation correspond to for current EHT targets?** This distinction is important because shadow resolvability does not necessarily imply metric distinguishability. A shadow may remain larger than the nominal instrumental resolution while the difference between competing spacetime geometries remains too small to identify reliably. A useful assessment therefore requires a direct

connection between the regularisation parameter, photon-sphere geometry, shadow diameter, and target-specific angular scale. This short communication addresses this question using a lightweight and reproducible computational framework for the non-rotating Bardeen metric. The photon-sphere radius and critical impact parameter are evaluated as functions of g , with the Schwarzschild limit providing the reference geometry. The resulting fractional shadow-diameter deviations are then converted into angular diameters for Sgr A* and M87* using their measured masses and distances and compared with the adopted EHT angular-resolution scale. Kerr shadow silhouettes are included separately to illustrate the effect of spin and distinguish its characteristic asymmetry from the size change associated with Bardeen regularisation. The contribution is therefore a **reproducible parameter-to-observable assessment** of Bardeen regularisation rather than a new black-hole metric or shadow formalism. The analysis provides a transparent computational baseline for evaluating how regularisation changes shadow size and for future studies incorporating spin, plasma effects, scattering, accretion-flow physics, and detailed EHT image modelling. The complete workflow is designed for execution in a browser-based Google Colab environment using standard scientific Python libraries.

3. Methods

3.1 Kerr shadow: analytic celestial coordinates

The Kerr shadow boundary was computed using the analytic formalism of Bardeen, Press and Teukolsky (1972) for unstable spherical photon orbits. The dimensionless spin parameter is defined as

$$a_* = \frac{a}{M},$$

where a is the Kerr spin parameter and M is the black-hole mass. For a given a_* , the allowed radial range of unstable spherical photon orbits was determined analytically. Each orbit was then mapped to the conserved quantities ξ and η , corresponding to the photon angular-momentum and Carter-constant parameters. For an observer at inclination i , these quantities were projected onto the celestial coordinates

$$\alpha = -\frac{\xi}{\sin i},$$

$$\beta = \pm \sqrt{\eta + a^2 \cos^2 i - \xi^2 \cot^2 i}.$$

The resulting (α, β) coordinates define the Kerr shadow boundary. In the non-rotating limit,

$$a_* = 0,$$

the shadow becomes circular with critical impact parameter

$$b_{\text{Sch}} = 3\sqrt{3}M,$$

and corresponding diameter

$$D_{\text{Sch}} = 2b_{\text{Sch}} = 6\sqrt{3}M.$$

This limit provides an internal consistency check for the numerical implementation. Representative Kerr silhouettes were generated for $a_* = 0, 0.5, 0.9$ to illustrate the effect of increasing spin on shadow asymmetry and displacement.

3.2 Bardeen regular black hole: metric, photon sphere, and shadow radius

The non-rotating Bardeen spacetime is described by the metric function introduced by Bardeen (1968) and later given a nonlinear-electrodynamic interpretation by Ayón-Beato and García (2000):

$$f(r) = 1 - \frac{2Mr^2}{(r^2 + g^2)^{3/2}},$$

where M is the black-hole mass and g is the regularisation parameter. The Schwarzschild limit is recovered as

$$g \rightarrow 0,$$

for which

$$f(r) \rightarrow 1 - \frac{2M}{r}.$$

For null geodesics in the static, spherically symmetric spacetime, the relevant impact-parameter function can be written as

$$h(r) = \frac{r^2}{f(r)}.$$

The photon-sphere radius r_{ph} is obtained from the extremum condition

$$\frac{dh(r)}{dr} = 0.$$

Because no convenient closed-form expression for r_{ph} is used here, the root of

$$\frac{d}{dr} \left[\frac{r^2}{f(r)} \right] = 0$$

is determined numerically using Brent's bracketed root-finding method over the selected radial interval. The corresponding critical impact parameter is

$$b_{\text{ph}} = \frac{r_{\text{ph}}}{\sqrt{f(r_{\text{ph}})}},$$

and the shadow diameter is therefore

$$D_{\text{sh}} = 2b_{\text{ph}}.$$

The fractional deviation from the Schwarzschild shadow is quantified as

$$\Delta D(\%) = \frac{D_{\text{Sch}} - D_{\text{sh}}}{D_{\text{Sch}}} \times 100.$$

The calculation is evaluated over the prescribed g/M parameter grid, allowing the photon-sphere radius, shadow diameter, and fractional deviation to be tracked continuously relative to the Schwarzschild reference.

3.3 Angular-scale conversion for Sagittarius A* and M87*

The dimensionless shadow diameter D_{sh}/M is converted into an observable angular diameter using the gravitational length scale

$$r_g = \frac{GM}{c^2}.$$

For a source at distance D , the corresponding physical shadow diameter is

$$L_{\text{sh}} = \left(\frac{D_{\text{sh}}}{M} \right) \frac{GM}{c^2},$$

and its angular diameter is

$$\theta_{\text{sh}} = \frac{L_{\text{sh}}}{D}.$$

In microarcseconds,

$$\theta_{\text{sh}}(\mu\text{as}) = \frac{L_{\text{sh}}}{D} \left(\frac{180}{\pi} \right) (3600)(10^6).$$

For Sagittarius A*, the adopted parameters are

$$M_{\text{Sgr A}^*} \approx 4.3 \times 10^6 M_{\odot}, \quad D_{\text{Sgr A}^*} \approx 8178 \text{ pc},$$

while for M87*,

$$M_{\text{M87}^*} \approx 6.5 \times 10^9 M_{\odot}, \quad D_{\text{M87}^*} \approx 16.8 \text{ Mpc}.$$

These values follow the EHT measurements used in the original analysis (Event Horizon Telescope Collaboration, 2019, 2022a). The calculated angular diameters are compared with the adopted EHT angular-resolution scale at 230 GHz. This comparison is used specifically to distinguish geometric resolvability from the substantially stronger requirement of parameter distinguishability. The latter also depends on calibration, systematic uncertainties, image reconstruction, and astrophysical effects not represented by the idealised metric calculation.

3.4 Reproducibility statement

The complete computational workflow including Kerr shadow generation, Bardeen photon-sphere determination, critical-impact-parameter calculation, angular-scale conversion, and figure generation—is implemented in a single self-contained Google Colab notebook. The calculations require only standard scientific Python libraries, including NumPy, SciPy, and Matplotlib, and can be executed on a CPU without specialised numerical-relativity software or GPU hardware.

The workflow follows the sequence

$$\{a_*, i\} \rightarrow (\alpha, \beta) \rightarrow \text{Kerr shadow},$$

and

$$\frac{g}{M} \rightarrow r_{\text{ph}} \rightarrow b_{\text{ph}} \rightarrow D_{\text{sh}} \rightarrow \theta_{\text{sh}},$$

with the resulting numerical quantities used directly to generate the tables and figures presented in this study. The browser-based implementation is intended to make the calculations readily reproducible and adaptable to other regular black-hole parameters, spacetime models, and observational targets.

4. Results

Figure 1 shows the main geometric differences between the Kerr and Bardeen cases. The Kerr shadows for $a_* = 0, 0.5$, and 0.9 exhibit the expected displacement and increasing asymmetry associated with rotation, whereas the Bardeen shadows for $g/M = 0, 0.3$, and 0.6 remain circular and become progressively smaller as g/M increases. This provides a clear qualitative distinction: Kerr spin primarily changes the shadow's shape and position, while Bardeen regularisation primarily changes its size.

The corresponding photon-sphere evolution is shown in Figure 3. The photon-sphere radius decreases smoothly from the Schwarzschild value,

$$r_{\text{ph}} = 3M,$$

at $g/M = 0$ to approximately

$$r_{\text{ph}} = 2.20M$$

at $g/M = 0.8$. The continuous decrease indicates that the reduction in shadow size is directly associated with the inward shift of the unstable photon orbit as the regularisation parameter increases.

The resulting shadow-diameter deviation is shown in Figure 4. The deviation is zero at

$$\frac{g}{M} = 0,$$

where the Bardeen metric exactly recovers the Schwarzschild limit, providing an internal check of the numerical calculation. The deviation then increases smoothly with g/M , reaching approximately

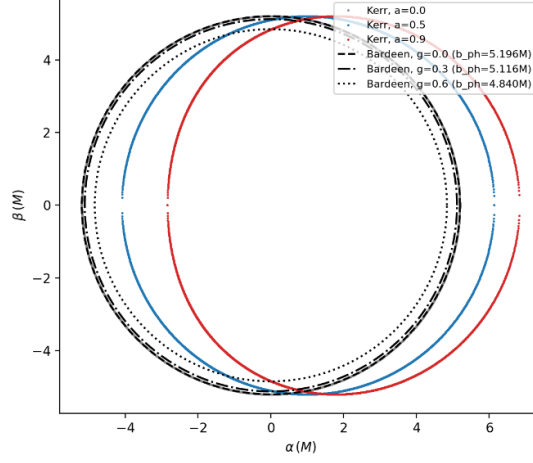
$$\Delta D \approx 1.51M$$

at $g/M = 0.8$. Relative to the Schwarzschild diameter,

$$D_{\text{Sch}} = 6\sqrt{3}M,$$

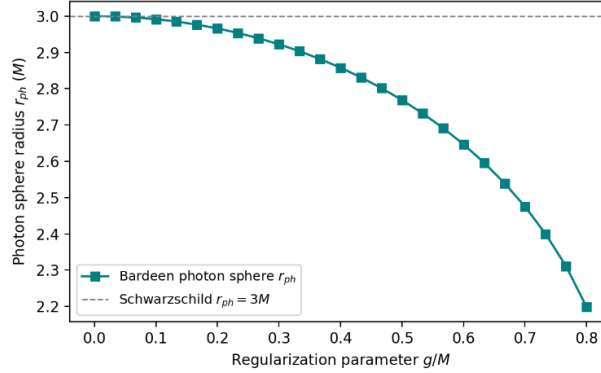
this corresponds to a fractional reduction of approximately 14.5%. The monotonic trend provides a direct numerical relationship between the Bardeen regularisation parameter and the observable shadow-size change

Figure 1: Shadow comparison — Kerr vs. Bardeen regular black hol

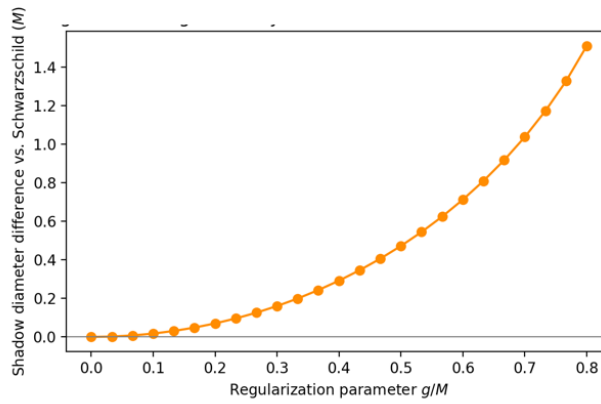


a

Figure 3: Photon sphere radius vs. regularization parameter



b



c

Figure 1. Geometric characterisation of the Bardeen–Kerr shadow comparison. (A) Shadow silhouettes in the observer’s sky plane. (B) Photon sphere radius contraction with increasing regularisation parameter g/M . (C)

Absolute shadow diameter deviation from the Schwarzschild reference, confirming exact recovery ($\Delta\text{diam} = 0$) at $g = 0$.

4. Results

Figure 2 places the geometric results in an observational context. The angular shadow diameter decreases progressively with increasing g/M for both Sagittarius A* and M87*. For Sagittarius A*, the predicted diameter decreases from approximately $53.9\mu\text{as}$ at $g/M = 0$ to $46.1\mu\text{as}$ at $g/M = 0.8$. For M87*, the corresponding decrease is from approximately $39.7\mu\text{as}$ to $34.2\mu\text{as}$. Both remain above the adopted EHT angular-resolution benchmark of approximately $22.5\mu\text{as}$ throughout the parameter range. Thus, the shadow itself remains geometrically resolvable; the more important issue is whether the change in shadow size is sufficiently large to distinguish the underlying spacetime. The fractional deviation is shown in Figure 5. It increases smoothly from zero at $g/M = 0$ to approximately 14.5% at $g/M = 0.8$. This provides a dimensionless measure of the effect that can be compared directly with observational uncertainties and systematic limitations.

Table 1. Photon-sphere radius, critical impact parameter, shadow diameter, and absolute deviation from the Schwarzschild reference as a function of the regularisation parameter g/M . The complete parameter grid is provided in the Supplementary Data.

g/M	$r_{\text{ph}} (M)$	$b_{\text{ph}} (M)$	Diameter (M)	$\Delta\text{diameter} (M)$
0.00	3.000	5.1962	10.3923	0.0000
0.10	2.987	5.1875	10.3749	0.0174
0.20	2.950	5.1611	10.3222	0.0701
0.30	2.885	5.1159	10.2319	0.1604
0.40	2.788	5.0501	10.1002	0.2922
0.50	2.653	4.9600	9.9201	0.4722
0.60	2.475	4.8397	9.6795	0.7128
0.70	2.243	4.6772	9.3544	1.0379
0.80	2.196	4.4409	8.8817	1.5106

The results in Table 1 show a smooth and monotonic reduction in both the photon-sphere radius and shadow diameter as g/M increases. At $g/M = 0$, the calculation recovers the Schwarzschild values,

$$r_{\text{ph}} = 3M,$$

$$b_{\text{ph}} = 3\sqrt{3}M,$$

and

$$D_{\text{Sch}} = 6\sqrt{3}M = 10.3923M,$$

giving

$$\Delta D = 0.$$

This exact recovery provides a useful numerical consistency check. At $g/M = 0.8$, the shadow diameter decreases to $8.8817M$, corresponding to an absolute reduction of $1.5106M$ and a fractional reduction of approximately 14.5%.

Taken together, Figures 2 and 5 show that increasing regularisation produces a measurable geometric change in the predicted shadow size while leaving the shadow above the adopted angular-resolution scale for both targets. The results therefore separate two quantities that should not be conflated: whether a shadow can be resolved and whether a deviation from the Schwarzschild/Kerr expectation can be statistically distinguished.

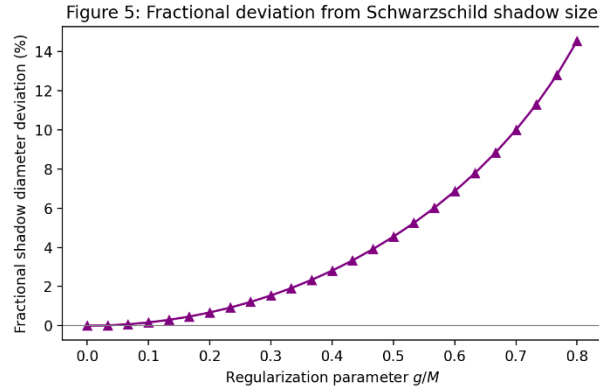
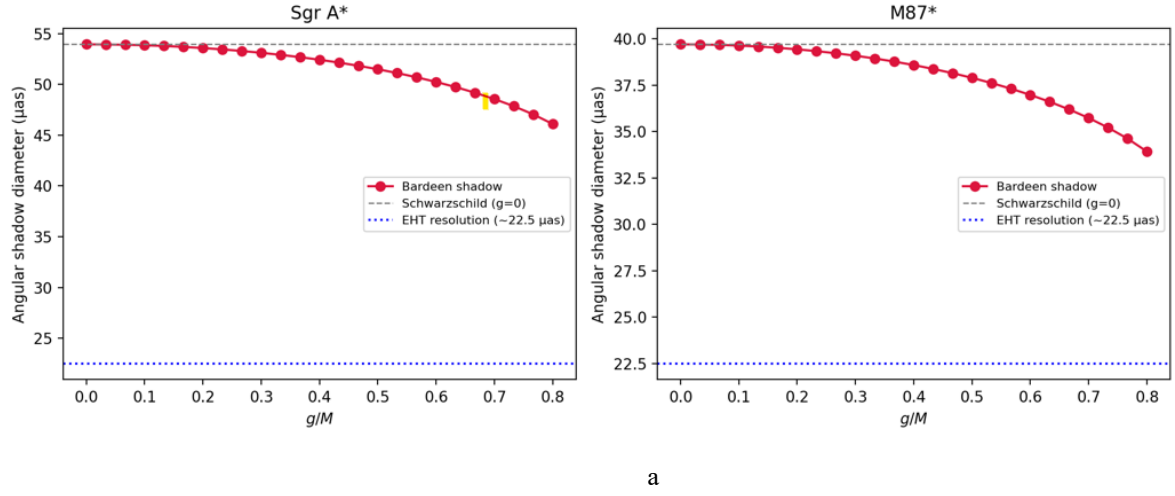


Figure 2. Observational relevance of the shadow deviation. (A) Angular shadow diameter for Sgr A* and M87*, compared to nominal EHT angular resolution ($\sim 22.5 \mu\text{as}$); both targets remain individually resolvable across the full g/M range tested. (B) Fractional shadow diameter deviation from the Schwarzschild case, reaching $\sim 14.5\%$ at $g = 0.8M$.

Table 2. Angular shadow diameter for Sagittarius A* and M87* at selected values of the Bardeen regularisation parameter g/M , compared with the adopted EHT angular-resolution scale of approximately $22.5 \mu\text{as}$ at 230 GHz.

Target	g/M	Diameter (M)	Angular diameter (μas)
Sgr A*	0.00 (Schwarzschild)	10.3923	53.941
Sgr A*	0.30	10.2319	53.108
Sgr A*	0.60	9.6795	50.241
M87*	0.00 (Schwarzschild)	10.3923	39.692
M87*	0.30	10.2319	39.079
M87*	0.60	9.6795	36.970

The values in Table 2 show a consistent reduction in angular shadow diameter as the regularisation parameter increases. For Sgr A*, the predicted diameter decreases from $53.941 \mu\text{as}$ at $g/M = 0$ to $50.241 \mu\text{as}$ at $g/M = 0.6$.

For M87*, the corresponding decrease is from $39.692\mu\text{as}$ to $36.970\mu\text{as}$. In both cases, the predicted shadow remains above the adopted $22.5\mu\text{as}$ angular-resolution benchmark. These results reinforce two separate points. First, Bardeen regularisation produces a smooth and quantitatively measurable reduction in shadow size. Second, the reduction does not prevent the shadow from remaining geometrically resolvable for either target under the adopted resolution benchmark. The magnitude of the deviation, rather than the basic visibility of the shadow, is therefore the more relevant quantity for assessing whether regularisation could be distinguished observationally. The corresponding fractional deviations are examined further in the Discussion.

5. Discussion

The results highlight an important distinction between shadow resolvability and shadow distinguishability. Both Sagittarius A* and M87* remain above the adopted EHT angular-resolution scale throughout the parameter range considered, including the largest value of g/M . Thus, the presence of a regularisation-induced change in the predicted angular scale does not, by itself, prevent the corresponding structure from being geometrically resolved. The more relevant question is whether the predicted change in shadow size is sufficiently large to be distinguished from observational, calibration, reconstruction, and astrophysical uncertainties. The fractional deviation from the Schwarzschild reference therefore provides a useful first-order measure of the magnitude of the geometric effect. The Bardeen prediction differs from the Schwarzschild reference by approximately 1.54%, 6.86%, and 14.53% for $g/M = 0.3, 0.6,$ and 0.8 , respectively. These values show that the effect is relatively modest at moderate regularisation but becomes progressively more pronounced as g/M increases. The resulting trend is therefore useful for identifying the parameter regime in which regularisation-induced changes become potentially relevant to horizon-scale observations. However, these percentages should be interpreted as geometric deviations of the model, rather than as direct observational exclusion or detection thresholds. The comparison with EHT observations should consequently be regarded as an order-of-magnitude benchmark rather than as a substitute for a full metric-testing analysis. In particular, the EHT analysis of Sagittarius A* incorporates calibration, image reconstruction, astrophysical modelling, and associated uncertainties when evaluating consistency with Kerr predictions (Event Horizon Telescope Collaboration, 2022b). A direct comparison between a calculated fractional shadow-size change and a single adopted uncertainty scale cannot reproduce this inference procedure. The present results therefore do not establish an observational constraint on the Bardeen parameter. Instead, they identify the range in which the predicted geometric effect becomes comparable to percent-level observational systematics and thereby motivate a more complete image-level analysis.

Horizon structure and the role of the $g/M = 0.8$ endpoint

A specific qualification is required for the upper end of the parameter range. The standard Bardeen solution possesses an extremal configuration at $g_{\text{ext}}/M \simeq 0.7698$, beyond which the event horizon disappears. Accordingly, $g/M = 0.8$ lies slightly beyond the horizon-bearing black-hole branch. Its inclusion in the present study is deliberate rather than accidental: it provides a controlled extension of the geometric parameter sweep across the extremal boundary and allows the response of the photon-orbit and optical critical-scale calculation to be examined immediately beyond the extremal limit. The $g/M = 0.8$ result should therefore not be interpreted as an ordinary horizon-bearing Bardeen black hole. More precisely, the value $8.8817M$ obtained at this endpoint represents the characteristic optical critical scale associated with the calculated unstable photon orbit under the adopted null-geodesic prescription. In the horizonless regime, it is preferable to regard this quantity as a formal shadow-like critical-curve scale rather than automatically assigning it the same physical interpretation as the black-hole shadow on the sub-extremal branch. This distinction preserves the continuity of the geometric calculation while avoiding the assumption that the over-extremal configuration possesses an event horizon. The choice of $g/M = 0.8$ is nevertheless informative because it probes the immediate neighbourhood beyond the extremal boundary rather than extending arbitrarily deep into the horizonless regime. The resulting continuation of the calculated optical scale provides a sensitivity test of how strongly the observable responds to increasing regularisation near and slightly beyond extremality. The approximately 14.53% reduction relative to the Schwarzschild reference should consequently be regarded as a formal upper sensitivity marker for the selected parameter sweep, rather than as an astrophysical prediction for a conventional Bardeen black hole. This distinction is particularly important when interpreting the endpoint in the context of EHT observations. The $g/M < g_{\text{ext}}/M$ cases correspond to the horizon-bearing branch of the Bardeen solution, whereas $g/M = 0.8$ is included as an over-extremal extension of the same metric prescription. The comparison therefore separates the physical black-hole regime from the formal geometric continuation and prevents the largest parameter value from being misinterpreted as evidence for an astrophysically realised horizonless object.

Limitations

Several additional limitations should be considered when interpreting the results:

- **Non-rotating assumption:** The analysis uses a spherically symmetric, non-rotating Bardeen metric. Astrophysical black holes are expected to possess spin, and a more complete treatment would require a rotating regular-black-hole geometry, such as the construction discussed by Bambi and Modesto (2013). A joint exploration of spin and regularisation would also allow possible degeneracies between rotation-induced shadow asymmetry and regularisation-induced size changes to be examined.
- **Vacuum geodesic optics:** The optical boundary is obtained from idealised vacuum null geodesics and does not include an accretion flow, plasma effects, or interstellar scattering. These environmental effects can modify the apparent size and morphology of black-hole images and have been considered in previous studies of regular black-hole shadows (Abdujabbarov et al., 2016; He et al., 2022). The present results should therefore be interpreted as geometric predictions of the underlying spacetime rather than direct predictions of reconstructed EHT images. This limitation is particularly relevant for the $g/M = 0.8$ extension, where the calculated optical scale should not automatically be equated with a conventional black-hole shadow.
- **Simplified observational baseline:** The comparison with EHT angular resolution and systematic uncertainty is intentionally simplified. It does not reproduce the EHT image-reconstruction process, calibration pipeline, astrophysical modelling, or complete uncertainty budget. Consequently, the reported fractional deviations should not be interpreted as observational constraints on the Bardeen parameter. Instead, they provide a quantitative scale against which future image-level analyses can be designed.
- **Extremality and parameter interpretation:** The inclusion of $g/M = 0.8$ extends the calculation slightly beyond the extremal black-hole threshold. This endpoint is retained specifically to test the behaviour of the geometric observable across the extremal transition. It is not used to expand the physically interpreted black-hole parameter space. The distinction between the horizon-bearing and horizonless branches is therefore maintained throughout the interpretation.

Future Work

These limitations also define the most direct directions for further work. A natural extension would combine spin and regularisation in a rotating regular-black-hole metric and evaluate the resulting shadow morphology over a two-parameter space. The geometric predictions could then be coupled to realistic accretion-flow and plasma models and compared with EHT image simulations and published metric-testing frameworks (Event Horizon Telescope Collaboration, 2022b). Such an approach would allow the size, asymmetry, brightness distribution, and other image-level observables to be considered simultaneously rather than relying primarily on the shadow diameter. Because the present workflow is lightweight and browser-based, it provides a practical starting point for these extensions without requiring specialised numerical-relativity infrastructure. Overall, the results establish a clear quantitative relationship between the Bardeen regularisation parameter and the predicted optical critical scale while demonstrating that geometric resolvability alone is insufficient to establish observational distinguishability. The sub-extremal results provide the physically relevant black-hole branch, while the $g/M = 0.8$ calculation serves as a controlled over-extremal sensitivity test of the same geometric framework. The analysis therefore provides a reproducible baseline for more complete tests of regular black-hole geometries using horizon-scale observations.

6. Conclusion

This study presents a lightweight and fully reproducible computational assessment of Bardeen regular black-hole shadow deviations using a single Google Colab notebook. Across the tested range $g/M = 0 - 0.8$, the photon-sphere radius decreases from $3.000M$ to $2.196M$, while the shadow diameter decreases from the Schwarzschild value of $10.3923M$ to $8.8817M$. The corresponding absolute diameter change reaches $1.5106M$, giving a maximum fractional deviation of approximately 14.5%. When converted to angular scales, the predicted shadow diameter decreases from $53.94\mu\text{as}$ to approximately $46.1\mu\text{as}$ for Sagittarius A* and from $39.69\mu\text{as}$ to approximately $34.2\mu\text{as}$ for M87* across the tested range. Both remain above the adopted EHT angular-resolution

benchmark of approximately $22.5\mu\text{s}$. Thus, the calculations indicate that Bardeen regularisation does not prevent geometric shadow resolution for either target, while the increasing fractional deviation provides a quantitative measure for assessing potential metric distinguishability. The results also show that the deviation is modest at intermediate regularisation values, reaching approximately 1.5% at $g/M = 0.3$ and 6.9% at $g/M = 0.6$. These values should be regarded as geometric benchmarks rather than direct observational constraints, since a rigorous EHT test requires image-level modelling that incorporates calibration, reconstruction, scattering, plasma, and accretion-flow effects. The analysis therefore identifies a quantitative parameter range for future observational studies rather than establishing a definitive detection threshold. Beyond the physical results, the complete workflow provides a transparent and reproducible computational baseline that can be executed in a free, browser-based environment. The framework can be extended to rotating regular black holes, joint spin-regularisation studies, realistic plasma and accretion models, and full EHT forward modelling, providing a practical starting point for more rigorous tests of regular black-hole geometries.

The end-to-end computational pipeline linking spacetime parameters to EHT observables is summarized in the graphical workflow (Figure A1)

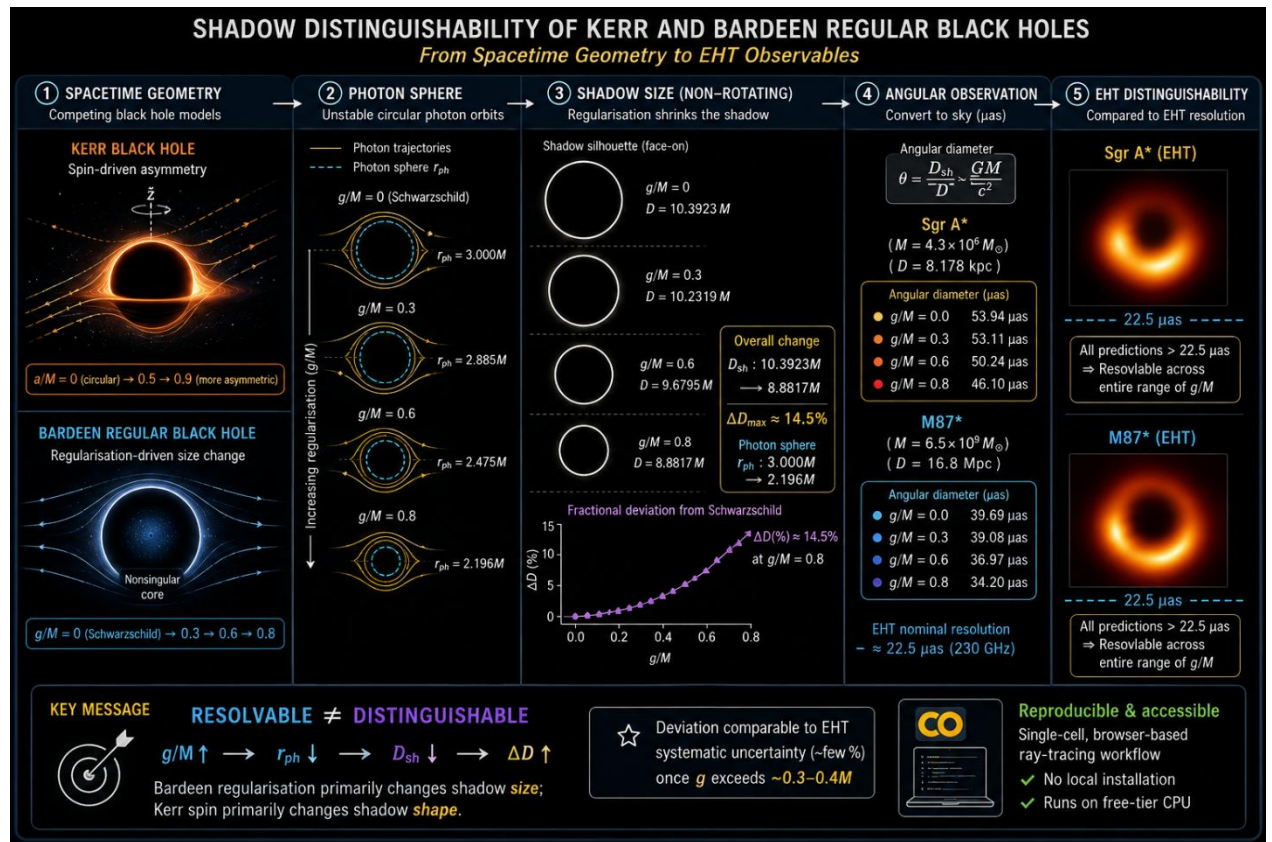


Figure A1. Graphical summary of the study. Schematic overview of the computational workflow linking Kerr and Bardeen black-hole geometries to photon-sphere properties, shadow-size deviations, and EHT-scale observables. Kerr spin primarily produces shadow asymmetry, whereas increasing Bardeen regularisation g/M progressively reduces the photon-sphere radius and shadow diameter. The analysis gives a maximum shadow-diameter deviation of approximately 14.5% at $g/M = 0.8$, while the predicted angular diameters for Sgr A* and M87* remain above the adopted EHT angular-resolution benchmark of $\sim 22.5\mu\text{s}$. The workflow is implemented as a reproducible, browser-based computational pipeline.

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Conflict of Interest Declaration

Conflict of Interest: The author declares that there are no conflicts of interest that could have influenced the research, analysis, or preparation of this manuscript.

AI-Assisted Writing Declaration

Use of AI: The author used **Grammarly** solely for checking grammar, spelling, punctuation, and general language clarity. No generative AI tool was used to generate the scientific content, analysis, results, interpretations, or conclusions of the manuscript. The author takes full responsibility for the content and final version of the manuscript.

References

1. Bardeen, J. M. (1968). Non-singular general-relativistic gravitational collapse. In *Proceedings of GR5* (International Conference on Gravitation and the Theory of Relativity), Tbilisi, USSR, p. 174.
2. Ayón-Beato, E., & García, A. (1998). Regular black hole in general relativity coupled to nonlinear electrodynamics. *Physical Review Letters*, 80, 5056–5059. arXiv:gr-qc/9911046.
3. Ayón-Beato, E., & García, A. (1999). New regular black hole solution from nonlinear electrodynamics. *Physics Letters B*, 464, 25–29. arXiv:hep-th/9911174.
4. Ayón-Beato, E., & García, A. (2000). The Bardeen model as a nonlinear magnetic monopole. *Physics Letters B*, 493, 149–152. arXiv:gr-qc/0009077.
5. Bardeen, J. M., Press, W. H., & Teukolsky, S. A. (1972). Rotating black holes: locally nonrotating frames, energy extraction, and scalar synchrotron radiation. *The Astrophysical Journal*, 178, 347–369.
6. Bambi, C., & Modesto, L. (2013). Rotating regular black holes. *Physics Letters B*, 721, 329–334. arXiv:1302.6075.
7. Li, Z., & Bambi, C. (2014). Measuring the Kerr spin parameter of regular black holes from their shadow. *Journal of Cosmology and Astroparticle Physics*, 2014(01), 041. arXiv:1309.1606.
8. Abdujabbarov, A., Amir, M., Ahmedov, B., & Ghosh, S. G. (2016). Shadow of rotating regular black holes. *Physical Review D*, 93, 104004. arXiv:1604.03809.
9. Stuchlík, Z., & Schee, J. (2019). Shadow of the regular Bardeen black holes and comparison of the motion of photons and neutrinos. *The European Physical Journal C*, 79, 44.
10. Rodrigues, M. E., & Silva, M. V. de S. (2018). Bardeen regular black hole with an electric source. *Journal of Cosmology and Astroparticle Physics*, 2018(06), 025. arXiv:1802.05095.
11. He, K.-J., Guo, S., Tan, S.-C., & Li, G.-P. (2022). Shadow images and observed luminosity of the Bardeen black hole surrounded by different accretions. *Chinese Physics C*, 46, 085106.
12. Event Horizon Telescope Collaboration. (2019). First M87 Event Horizon Telescope results. I. The shadow of the supermassive black hole. *The Astrophysical Journal Letters*, 875, L1.
13. Event Horizon Telescope Collaboration. (2022). First Sagittarius A* Event Horizon Telescope results. I. The shadow of the supermassive black hole in the center of the Milky Way. *The Astrophysical Journal Letters*, 930, L12.
14. Event Horizon Telescope Collaboration. (2022). First Sagittarius A* Event Horizon Telescope results. VI. Testing the black hole metric. *The Astrophysical Journal Letters*, 930, L17. arXiv:2311.09484.
15. Held, A., Gold, R., & Eichhorn, A. (2019). Asymptotic safety casts its shadow. *Journal of Cosmology and Astroparticle Physics*, 2019(06), 029.

Appendix A: Pseudocode for the Shadow Ray-Tracing Pipeline

The following pseudocode summarises the complete computational workflow described in Section 3. The implementation is designed to run in a single Google Colab notebook using standard scientific Python libraries.

Algorithm A1. Kerr Shadow Boundary from Analytic Celestial Coordinates

Input: Kerr spin parameter a , observer inclination θ_{obs} , black-hole mass M .

Output: Set of celestial coordinates (α, β) defining the shadow boundary.

1. Schwarzschild limit:

If $a = 0$, return the circular shadow

$$\alpha^2 + \beta^2 = b_{\text{Sch}}^2,$$

with $b_{\text{Sch}} = 3\sqrt{3}M$.

2. Spherical photon orbits:

For $a \neq 0$, determine the radial interval $r \in [r_{\text{min}}, r_{\text{max}}]$ corresponding to unstable spherical photon orbits using the analytic relations of Bardeen, Press and Teukolsky (1972).

3. Conserved quantities:

For each sampled radius r , calculate

$$\xi(r) = \frac{L_z}{E}, \quad \text{and} \quad \eta(r) = \frac{Q}{E^2},$$

where E , L_z , and Q are the photon energy, axial angular momentum, and Carter constant, respectively.

4. Physical-orbit condition:

If $\eta(r) < 0$, the corresponding orbit is discarded.

5. Celestial-coordinate projection:

For an observer at inclination θ_{obs} ,

$$\alpha = -\frac{\xi}{\sin \theta_{\text{obs}}},$$

and

$$\beta = \pm \sqrt{\eta + a^2 \cos^2 \theta_{\text{obs}} - \xi^2 \cot^2 \theta_{\text{obs}}}.$$

If $\beta^2 < 0$, the corresponding point is discarded.

6. Store: $(\alpha, +\beta)$ and $(\alpha, -\beta)$ for every physically allowed orbit.

7. Return: The complete set $S_{\text{Kerr}} = \{(\alpha_i, \beta_i)\}$.

Algorithm A2. Bardeen Regular Black Hole: Photon Sphere and Shadow Radius

Input: Regularisation parameter g , black-hole mass M .

Output: Photon-sphere radius r_{ph} and critical impact parameter b_{ph} .

1. The Bardeen metric function is defined as

$$f(r) = 1 - \frac{2Mr^2}{(r^2 + g^2)^{3/2}}.$$

2. The effective impact-parameter function is

$$h(r) = \frac{r^2}{f(r)}.$$

3. The photon-sphere radius satisfies $\frac{dh(r)}{dr} = 0$, or equivalently,

$$\frac{d}{dr} \left[\frac{r^2}{f(r)} \right] = 0.$$

4. **Define the radial search interval:** $r \in [2.01M, 6M]$.
5. Evaluate $\frac{dh}{dr}$ over the search interval and identify a bracket containing the relevant root.
6. Apply Brent's bracketed root-finding method to obtain r_{ph} .
7. **Evaluate the metric function at the photon sphere:** $f_{\text{ph}} = f(r_{\text{ph}})$.
8. **Calculate the critical impact parameter:**

$$b_{\text{ph}} = \frac{r_{\text{ph}}}{\sqrt{f(r_{\text{ph}})}}.$$

9. **Calculate the shadow diameter:** $D_{\text{sh}} = 2b_{\text{ph}}$.
10. **Return:** $(r_{\text{ph}}, b_{\text{ph}}, D_{\text{sh}})$.

Algorithm A3. Shadow-Distinguishability Metrics

Input: Grid of regularisation parameters g/M , Schwarzschild reference diameter.

1. The Schwarzschild reference is

$$D_{\text{Sch}} = 2b_{\text{Sch}} = 6\sqrt{3}M \approx 10.3923M.$$

2. For each value of g/M :

- Evaluate Algorithm A2: $(r_{\text{ph}}, b_{\text{ph}}) = A_2(g, M)$.
- Calculate the Bardeen shadow diameter: $D_{\text{sh}} = 2b_{\text{ph}}$.
- Calculate the absolute diameter deviation: $\Delta D = D_{\text{Sch}} - D_{\text{sh}}$.
- Calculate the fractional diameter deviation:

$$\Delta D(\%) = \frac{D_{\text{Sch}} - D_{\text{sh}}}{D_{\text{Sch}}} \times 100.$$

- Store $\left(\frac{g}{M}, r_{\text{ph}}, b_{\text{ph}}, D_{\text{sh}}, \Delta D, \Delta D(\%) \right)$.

3. **Perform the Schwarzschild recovery check:**

$$\Delta D|_{g=0} = 0, \quad \text{and} \quad D_{\text{sh}}|_{g=0} = 6\sqrt{3}M.$$

The resulting dataset provides the numerical basis for Table 1 and the shadow-deviation figures.

Algorithm A4. Angular-Scale Conversion and EHT Comparison

Input: Dimensionless shadow diameter D_{sh}/M , black-hole mass $M_{\odot}$, source distance D_{pc} .

Output: Angular shadow diameter θ_{sh} in microarcseconds.

1. The gravitational length scale is

$$r_g = \frac{GM_{\text{BH}}}{c^2},$$

where $M_{\text{BH}} = M_{\odot} \left(\frac{M_{\text{BH}}}{M_{\odot}} \right)$.

2. The physical shadow diameter is

$$L_{\text{sh}} = \left(\frac{D_{\text{sh}}}{M} \right) r_g.$$

3. The source distance is converted to metres according to $D_m = D_{\text{pc}} \times (1 \text{ pc})$.
4. For the small-angle approximation,

$$\theta_{\text{sh, rad}} = \frac{L_{\text{sh}}}{D_m}.$$

5. Conversion to microarcseconds gives

$$\theta_{\text{sh}}(\mu\text{as}) = \theta_{\text{sh, rad}} \left(\frac{180}{\pi} \right) (3600)(10^6).$$

6. The calculation is applied to Sgr A* and M87* using the adopted masses and distances from Event Horizon Telescope Collaboration (2019, 2022a).
7. The resulting angular diameter is then compared with the adopted EHT angular-resolution benchmark, $\theta_{\text{EHT}} \approx 22.5 \mu\text{as}$, to determine whether $\theta_{\text{sh}} > \theta_{\text{EHT}}$.

This comparison represents geometric resolvability and is not treated as a direct statistical test of metric distinguishability.

Algorithm A5. Complete Computational Workflow

The complete pipeline is executed sequentially as follows:

$$(a, \theta_{\text{obs}}, M) \rightarrow (\xi, \eta) \rightarrow (\alpha, \beta) \rightarrow S_{\text{Kerr}}$$

for the Kerr calculations, and

$$\frac{g}{M} \rightarrow r_{\text{ph}} \rightarrow b_{\text{ph}} \rightarrow D_{\text{sh}} \rightarrow \Delta D \rightarrow \Delta D(\%)$$

for the Bardeen calculations.

The end-to-end procedure is:

1. Generate Kerr shadows for $a/M = 0, 0.5, 0.9$, providing the Kerr silhouettes.
2. Evaluate the Bardeen photon sphere and shadow for representative values $g/M = 0, 0.3, 0.6$.
3. Evaluate Algorithm A3 over $0 \leq \frac{g}{M} \leq 0.8$, providing the photon-sphere evolution, shadow diameter, absolute deviation, and fractional deviation.
4. Apply Algorithm A4 to Sgr A* and M87* over the same g/M grid.
5. Generate the corresponding figures and numerical tables.
6. Export the figures and numerical results from the Colab environment.

The complete computational sequence can therefore be summarised as

$$\frac{g}{M} \rightarrow r_{\text{ph}} \rightarrow b_{\text{ph}} \rightarrow D_{\text{sh}} \rightarrow \Delta D \rightarrow \Delta D(\%) \rightarrow \theta_{\text{sh}}$$

which provides a direct and reproducible mapping from the Bardeen regularisation parameter to the predicted shadow observable.